FLAME AND ICICLE GRAPHS

Brendan Gregg, performance maestro of Netflix fame, invented flame graphs (https://www.brendangregg.com/flamegraphs.html) when, while working on a MySQL performance issue, he needed to understand CPU usage. Brendan released them in late 2011, and the core idea of this visualization (figure 9.2) is as follows:

- The stack profiles are located on the x-axis. Note that the x-axis *does not* represent time but the collection of stack traces. Each rectangle represents a stack trace (compare call stacks in programming languages) or, put more simply, a call to a function or operating system syscall. The width of the rectangle represents the number of times the function was called or how much time it spent running. For example, we see in figure 9.2 that `slowTask()` spent much more time on CPU than `quickTask()`.
- The y-axis shows the stack depth, counting from zero (function call entry point). The rectangle at the top represents the function call that is on-CPU (that is, actually running) at the point of stack trace collection. In our example, we can see that `slowTask()` was called from `main()`.
- Originally, Brendan used random colors to help differentiate stack frames; however, they could be used to convey additional information. In other words, the colors do not have semantics, per se, and are only meant to help you navigate the flame graph.

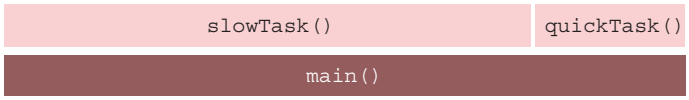


Figure 9.2 An flame graph using our `pprof` Go example

While flame graphs for resources were originally focused on CPU usage, they have grown into covering a wider set of resources, including main memory (RAM) or network I/O as well as runtime and the operating system level (e.g., garbage collection in a Java virtual machine or Go).

Having said that, don't think of flame graphs as being restricted to system-level tasks. In fact, being able to see both what your operating system (or JVM) does as well as how your application is performing with respect to execution time or CPU and memory usage is the real sweet spot. You may also come across the term *icicle charts*, which simply refers to upside-down flame graphs (as an icicle would grow from the top to the bottom). If you want to dive deeper into the topic, I recommend reading Polar Signal's article, "Ice and Fire: How to Read Icicle and Flame Graphs" (http://mng.bz/2DP0).